Asrafuzzaman Khan Nahin

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Education

M.S. in Electrical Engineering (Electronics) Jan 2024 - May 2026
King Fahd University of Petroleum and Minerals (KFUPM)

- CGPA: 3.625/4.00
- Supervisor: Dr. Jhonathan Prieto Rojas
- Research focused on PCB autorouting, Electronic Design Automation (EDA), optimization algorithms, and machine learning-assisted routing.
- Tools: Cadence, KiCad, ANSYS, MATLAB, Python, C++.

B.Sc. in Electrical and Electronic Engineering Jan 2019 - May 2023
Islamic University of Technology (IUT)

- CGPA: 3.49/4.00
- Supervisor: Mr. Md. Omar Faruque
- Specialized in photonics, electronics, and communication engineering.
- Tools: ANSYS Lumerical, COMSOL Multiphysics, MATLAB, PSpice.

Projects

Flow-field Based Routing Algorithms for Efficient PCB Design Automation Jan 2024 - May 2026
King Fahd University of Petroleum and Minerals (KFUPM)

- Developed a scalable 3D flow-field-based PCB autorouting framework integrating Dijkstra propagation, vector-field guidance, and via-aware cost modeling for multi-layer routing.
- Integrated a U-Net-based routing-region prediction model trained on the PCBench dataset to guide routing decisions and enhance global routability.
- Implemented and evaluated the framework in Python and C++, demonstrating improved runtime efficiency and competitive routing performance on large-scale PCB benchmarks.

Design of a High-efficiency Gold Bottom Reflector Based Grating Coupler with Uneven Grating Dec 2022 - May 2023
Islamic University of Technology (IUT)

- Optimized high-efficiency grating coupler design, boosting light coupling efficiency from 67% to 89.5%.
- Utilized ANSYS tools for computer simulations and design explorations of grating couplers.

Technical Skills

Programming	MATLAB, C/C++, Python, Verilog
Machine Learning	PyTorch
Professional Software	Simulink, ANSYS Lumerical/HFSS, COMSOL, Cadence, KiCad
Drawing & Typesetting	MS Office, LaTeX, Adobe Illustrator
Languages	English, Bengali (Native)

Publications

An Overview of Computer-Aided Printed Circuit Board Routing in the Era of Next-Generation Intelligent Autorouting Techniques. *Under Review.*

A. K. Nahin, A. A. R. Akib, R. Al Mahmud, and K. K. Qureshi, *"Simulation-Based Design and Optimization of a Plasmonic Metamaterial Absorber for Long-Wave Infrared Microbolometers," Arab J Sci Eng*, Jul. 2026, doi: 10.1007/s13369-026-11485-7.

T. Afdol et al., *"Numerical analysis of an asymmetric hexagonal plasmonic refractive index sensor model based on metal–insulator–metal and Si-insulator-Si waveguide," Sensing and Bio-Sensing Research*, vol. 40, p. 100563, Jun. 2023, doi: 10.1016/j.sbsr.2023.100563.

Professional Experience

Teaching Assistant, King Fahd University of Petroleum and Minerals Jan 2024 - May 2026

– Assisted course instruction and grading for Industrial Electronics and Bioelectronics courses.

– Supervised the Industrial Electronics Lab, including experiment setup and student troubleshooting support.

Extracurricular Activities

– Taught physics at Science Coaching, a center for students from various schools, for 5 months in 2019.

– Contributed to the Mars Rover electronics team, developing arm-control code and building key circuits, and participated in multiple International Rover Design Challenges.

Awards and Scholarships

– **OIC Scholarship** 2019

– **KFUPM MS Scholarship** 2024

– **KFUPM PhD Scholarship** 2026

References

Dr. Jhonathan Prieto Rojas
Assistant Professor
King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia
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Dr. Rabiul Al Mahmud
Assistant Professor
Islamic University of Technology (IUT), Gazipur, Bangladesh
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